How to Size a Rainwater Collection System

Posted By

Ashley Cofer on June 01, 2026



Find the right balance
between storage capacity,
cost, and water independence.

When you decide to collect
rainwater, one of the first
questions is:

**How big should my
rainwater system be?**

Sizing a rainwater harvesting
system is about balancing
three things:

How much rain you receive

How much water you use

How much storage you need
during dry periods
Here's a simple step-by-step
guide to help you estimate the
right system size for your
home, farm, or homestead.

Measure Your Catchment Area

Your roof is your water collection surface.

Start by measuring the square footage of the roof areas that drain into your gutters.

Formula

Catchment Area (sq ft)} = Length x Width

Only count the roof sections that actually feed your storage tank.

For example, a home with a 2,000-square-foot roof may provide roughly 2,000 square feet of usable catchment area.

Know Your Local Rainfall

Next, determine your average annual rainfall.

You can find this information through the NOAA Climate Data Center.

For example:

Greenville, SC receives about 50 inches of rainfall annually, which is excellent for rainwater harvesting.

Calculate Potential Water Collection

Use this formula to estimate how much water your roof can collect each year:

Gallons = 0.623 x Rainfall (in) x Catchment Area (sq ft) x Runoff Coefficient

What the Numbers Mean

0.623 = gallons collected per square foot per inch of rain

Runoff Coefficient = efficiency based on roof material
Typical runoff coefficients:

Roof	Coef

Type	ficient
Metal Roof	0.95
Asphalt Shingl	0.85

e s	
T i l e R o o f	0 . 7 5

Example Calculation

A 2,000 sq ft metal roof in a 50-inch rainfall area:

2,000 x 50 x 0.623 x 0.95
approx. 59,000 gallons/year

That’s nearly 5,000 gallons per month from a single roof.

Estimate Your Water Usage

Now determine what the rainwater system will supply.

Use Type	Average Demand
Garden	5–10 gal

rrigation	lons/dayper100sqft
Laundryandtoil	10–20gallons/p

ets	erson/day
Whole-house potable use	40–60 gallons/person/d

	a y

Add up your estimated monthly demand and compare it to your expected rainfall collection.

Choose the Right Tank Size

Your storage tank helps bridge periods without rain.

In much of the Southeast, a good rule of thumb is to store about 2 to 2.5 months of expected water demand.

Example

If your household uses:

- 3,000 gallons per month

A good tank size would typically be:

- 6,000–7,500 gallons

Areas with longer dry seasons may require larger storage capacity.

Plan for Overflow and First Flush

Every rainwater system should include:

Overflow Protection

An overflow line safely directs excess water away from foundations and structures when tanks are full.

First-Flush Diverter

A first-flush system discards the initial roof runoff during each rain event. This helps remove:

- Dust
- Pollen
- Bird droppings
- Roof debris

before water enters the tank.

Filter and Treat the Water

Non-Potable Systems

For irrigation, laundry, or toilet use:

Leaf screens

Storage tank
are often sufficient.

Potable Systems

For drinking water applications, additional treatment is typically required:

Carbon filtration

5 micro sediment filter

UV sterilization
Always follow local and state health regulations for potable water systems.

The Takeaway

Sizing a rainwater harvesting system is all about balance:

Collect enough water to meet your needs

Store enough to handle dry periods

Filter enough to maintain water quality and safety
A properly designed rainwater system can provide:

Greater water independence

Backup supply during outages or drought

Reduced reliance on chemically treated municipal water

Long-term resilience for homes, farms, and gardens

Ready to Get Started?

A rainwater harvesting system can be tailored to your:

rainfall levels

roof size

water usage

property goals

Whether you want a small garden system or a full household water supply, proper sizing is the key to reliable performance.